

Aditya ML 1

```
## ML LAB 1
### NAME : Aditya Bhende
### SECTION-BATCH : C-C2
### ROLL NO. : 28
```

In [2]

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

# from sklearn.preprocessing import LabelEncoder
# from sklearn.preprocessing import StandardScaler

# from sklearn.model_selection import train_test_split
```

In [3]

```
df = pd.read_csv("titanic.csv")
```

In [4]

```
df.head()
```

Output:

```
   PassengerId  Survived  Pclass \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

   Name                               Sex  Age  SibSp \
0   Braund, Mr. Owen Harris          male  22.0     1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0     1
2   Heikkinen, Miss. Laina          female  26.0     0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0     1
4    Allen, Mr. William Henry          male  35.0     0

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0   113803   53.1000  C123      S
4      0   373450   8.0500   NaN      S
```

In [5]

```
df.tail()
```

Output:

```
   PassengerId  Survived  Pclass                               Name \
886          887         0       2   Montvila, Rev. Juozas
887          888         1       1   Graham, Miss. Margaret Edith
888          889         0       3  Johnston, Miss. Catherine Helen "Carrie"
889          890         1       1   Behr, Mr. Karl Howell
890          891         0       3   Dooley, Mr. Patrick

   Sex  Age  SibSp  Parch  Ticket   Fare Cabin Embarked
886  male  27.0    0      0   211536  13.00   NaN      S
887  female  19.0    0      0   112053  30.00  B42      S
888  female   NaN    1      2  W./C. 6607  23.45   NaN      S
```

```
889    male  26.0      0      0      111369  30.00  C148      C
890    male  32.0      0      0      370376   7.75   NaN      Q
```

In [6]

```
df.shape
```

Output:

```
(891, 12)
```

In [7]

```
df.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [8]

```
df.describe()
```

Output:

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

In [9]

```
df.isnull().sum()
```

Output:

```
PassengerId    0
Survived        0
```

```
Pclass      0
Name        0
Sex         0
Age        177
SibSp       0
Parch       0
Ticket      0
Fare        0
Cabin      687
Embarked     2
dtype: int64
```

In [10]

```
df["Embarked"].unique()
```

Output:

```
array(['S', 'C', 'Q', nan], dtype=object)
```

In [11]

```
df["Embarked"] = df["Embarked"].replace("S",0)
df["Embarked"] = df["Embarked"].replace("C",1)
df["Embarked"] = df["Embarked"].replace("Q",2)
```

Output:

```
C:\Users\HP\AppData\Local\Temp\ipykernel_14060\1861778179.py:3: FutureWarning: Downcasting behavior in `replace` is deprecated
df["Embarked"] = df["Embarked"].replace("Q",2)
```

In [12]

```
df.head()
```

Output:

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	0.0
1	0	PC 17599	71.2833	C85	1.0
2	0	STON/O2. 3101282	7.9250	NaN	0.0
3	0	113803	53.1000	C123	0.0
4	0	373450	8.0500	NaN	0.0

In [13]

```
df["Sex"] = df["Sex"].replace("male",0)
df["Sex"] = df["Sex"].replace("female",1)
```

Output:

```
C:\Users\HP\AppData\Local\Temp\ipykernel_14060\1826372962.py:2: FutureWarning: Downcasting behavior in `replace` is deprecated
df["Sex"] = df["Sex"].replace("female",1)
```

In [14]

```
df.head()
```

Output:

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	Parch	\
0	Braund, Mr. Owen Harris	0	22.0	1	0	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	1	38.0	1	0	
2	Heikkinen, Miss. Laina	1	26.0	0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	1	35.0	1	0	
4	Allen, Mr. William Henry	0	35.0	0	0	

	Ticket	Fare	Cabin	Embarked
0	A/5 21171	7.2500	NaN	0.0
1	PC 17599	71.2833	C85	1.0
2	STON/O2. 3101282	7.9250	NaN	0.0
3	113803	53.1000	C123	0.0
4	373450	8.0500	NaN	0.0

In [15]

```
df["Age"].fillna(df["Age"].median(), inplace = True)
```

Output:

C:\Users\HP\AppData\Local\Temp\ipykernel_14060\687817444.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series, and its current value is nan. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are operating is a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col].method(value, inplace=True) instead.

```
df["Age"].fillna(df["Age"].median(), inplace = True)
```

In [16]

```
df["Embarked"].fillna(df["Embarked"].mode()[0],inplace=True)
```

Output:

C:\Users\HP\AppData\Local\Temp\ipykernel_14060\548196730.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series, and its current value is nan. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are operating is a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col].method(value, inplace=True) instead.

```
df["Embarked"].fillna(df["Embarked"].mode()[0],inplace=True)
```

In [17]

```
df.isnull().sum()
```

Output:

```
PassengerId    0
Survived        0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         687
Embarked        0
dtype: int64
```

In [18]

```
df.drop("Cabin", axis=1, inplace=True)
```

In [19]

```
df.drop("PassengerId", axis=1, inplace=True)
```

In [20]

```
df.drop("Ticket", axis=1, inplace=True)
```

In [21]

```
df.drop("Name", axis=1, inplace=True)
```

In [22]

```
df.duplicated().sum()
```

Output:

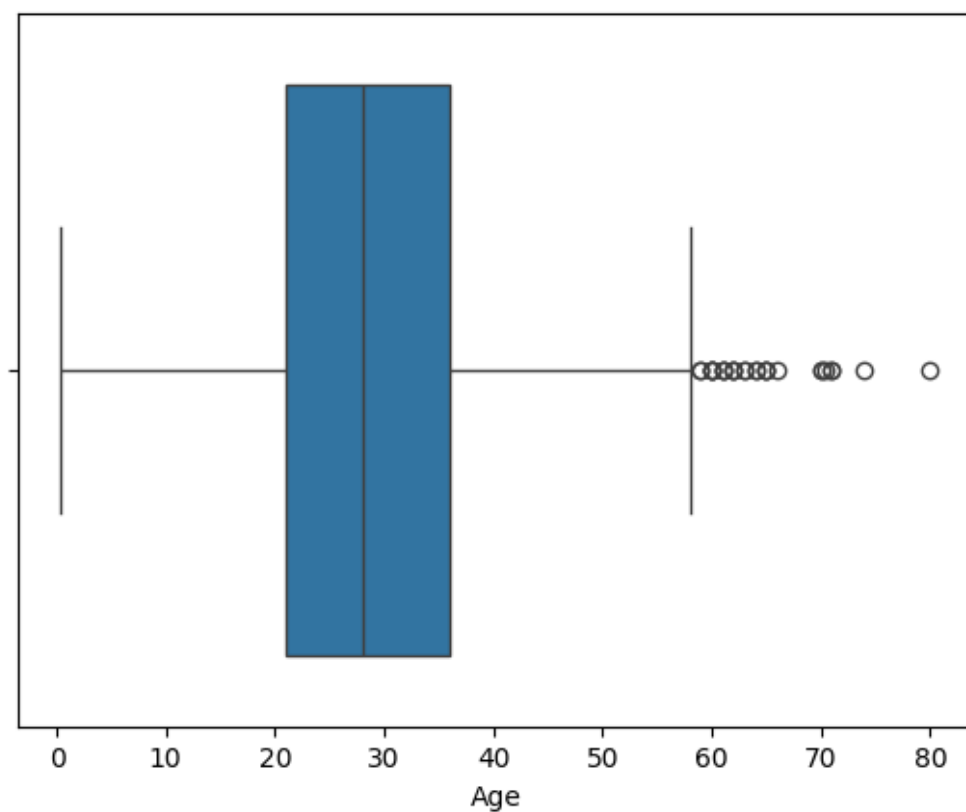
```
116
```

In [23]

```
df.drop_duplicates(inplace=True)
```

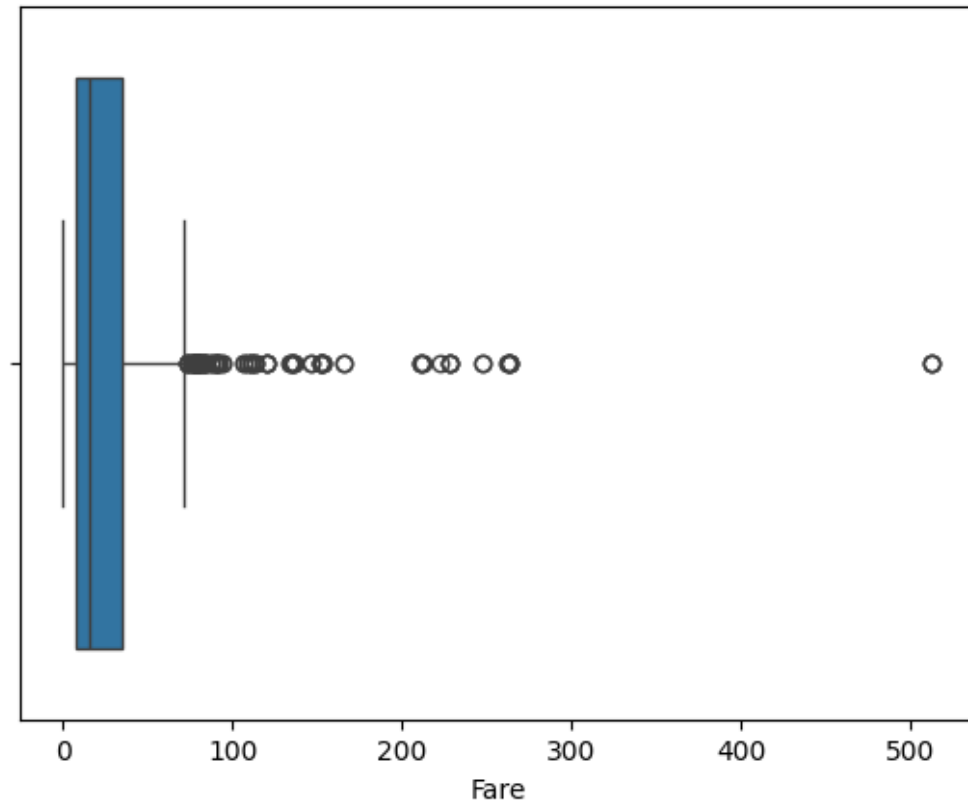
In [24]

```
sns.boxplot(x=df["Age"])  
plt.show()
```



In [25]

```
sns.boxplot(x=df["Fare"])  
plt.show()
```

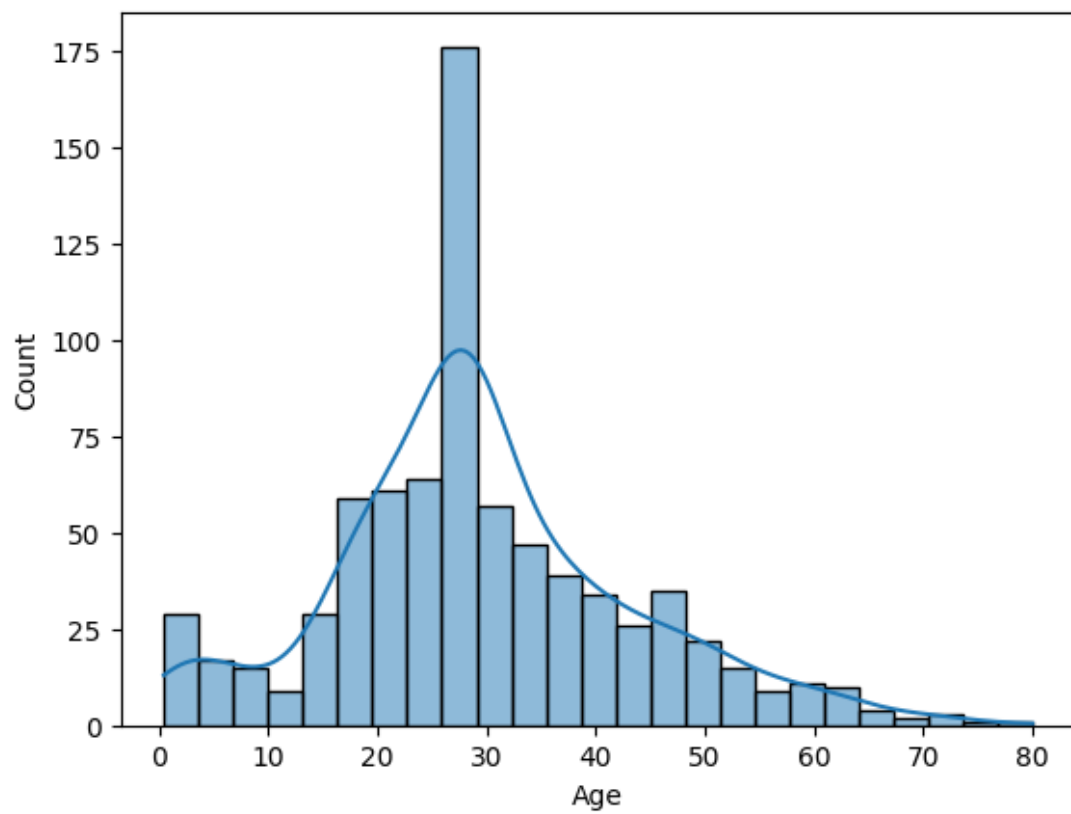


In [26]

```
df=pd.get_dummies(df, columns=["Embarked"])
```

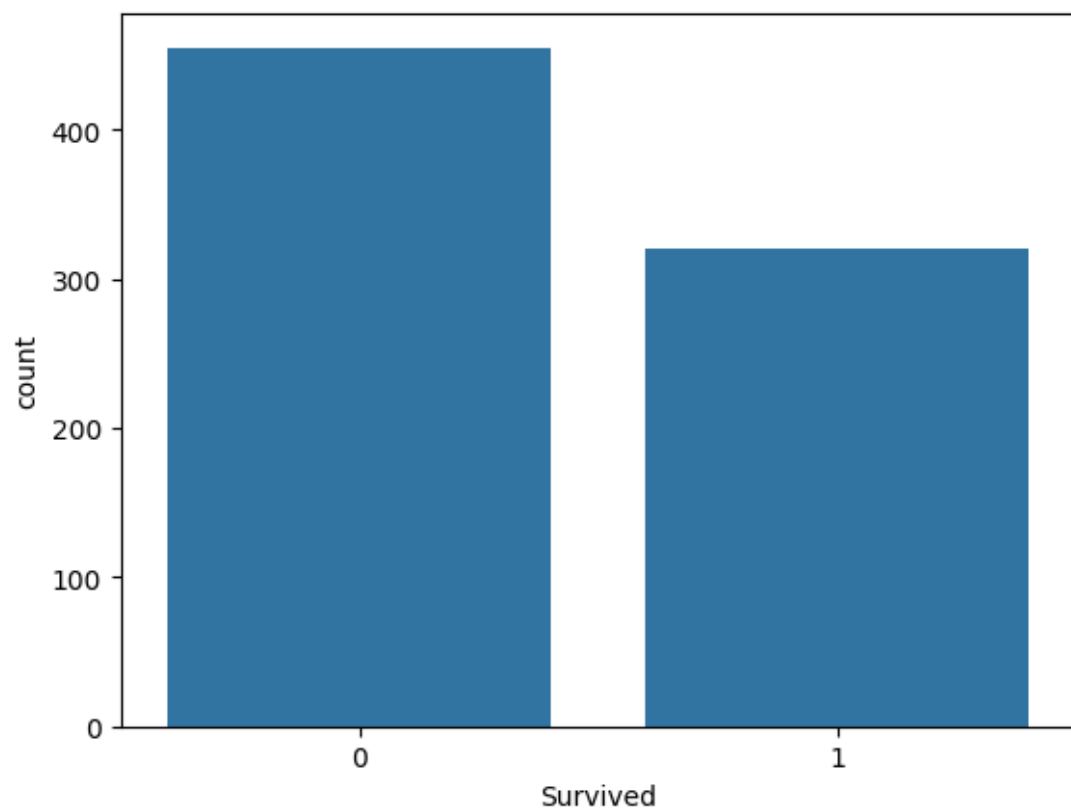
In [27]

```
sns.histplot(df["Age"], kde=True)  
plt.show()
```



In [28]

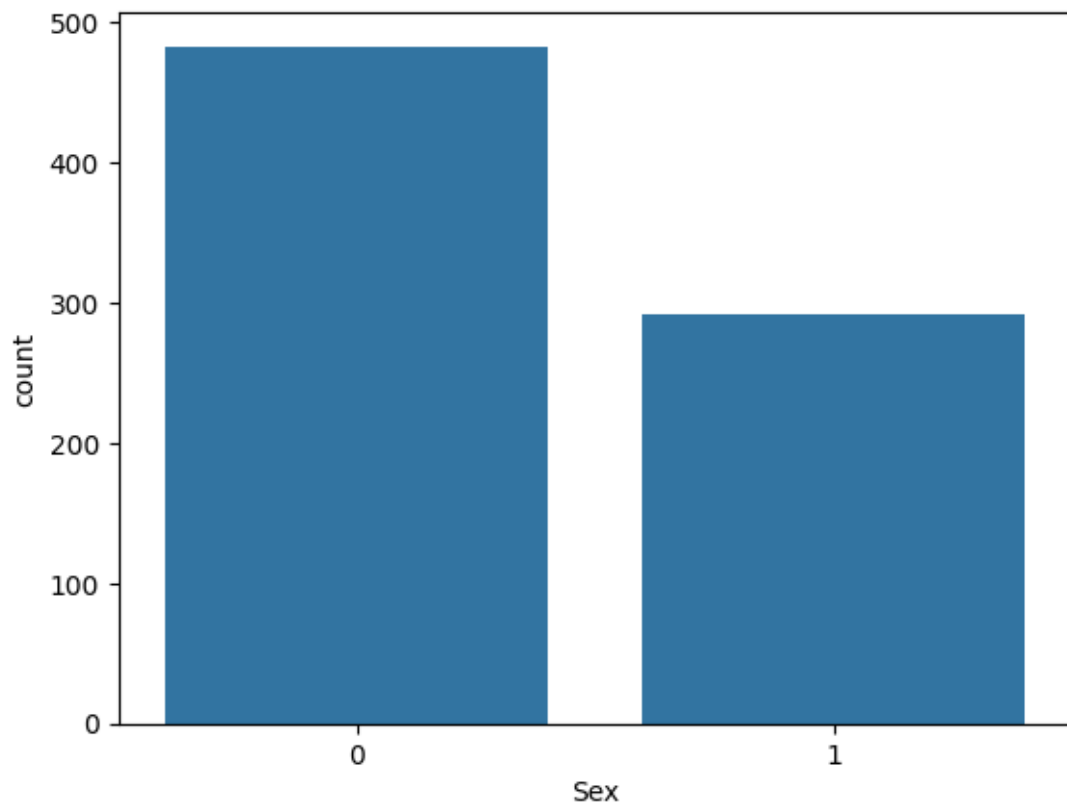
```
sns.countplot(x="Survived", data=df)  
plt.show()
```



In [29]

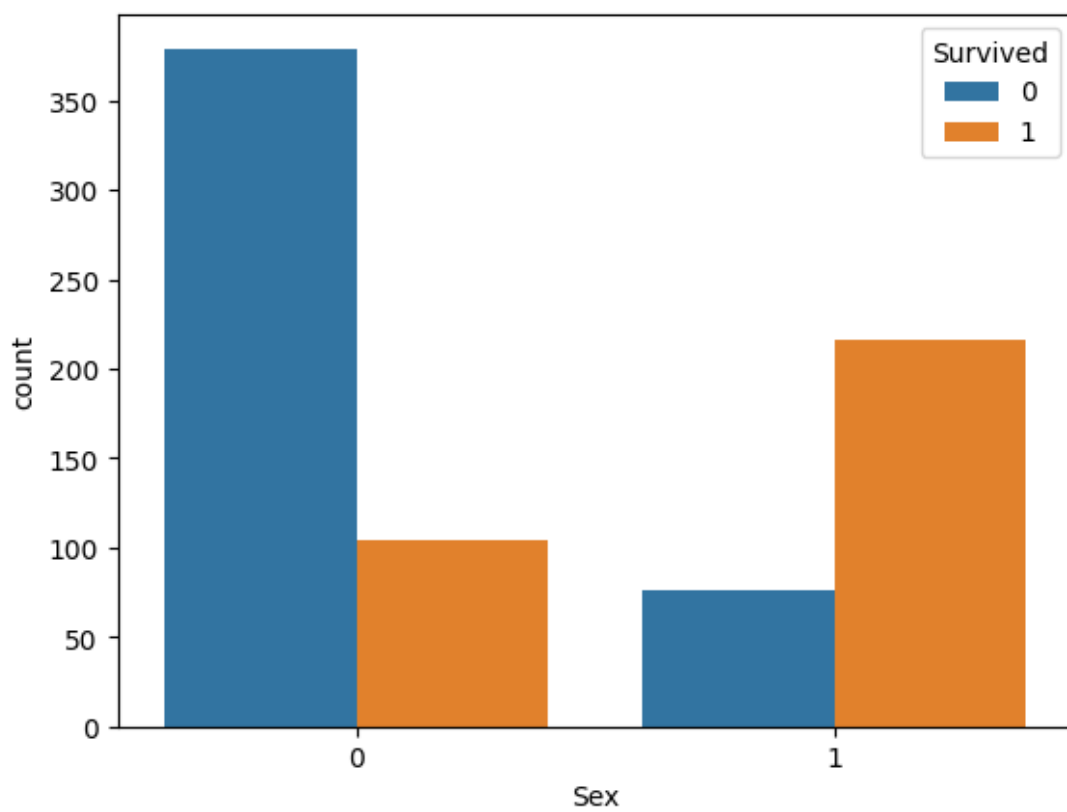
```
sns.countplot(x="Sex", data=df)
```

```
plt.show()
```



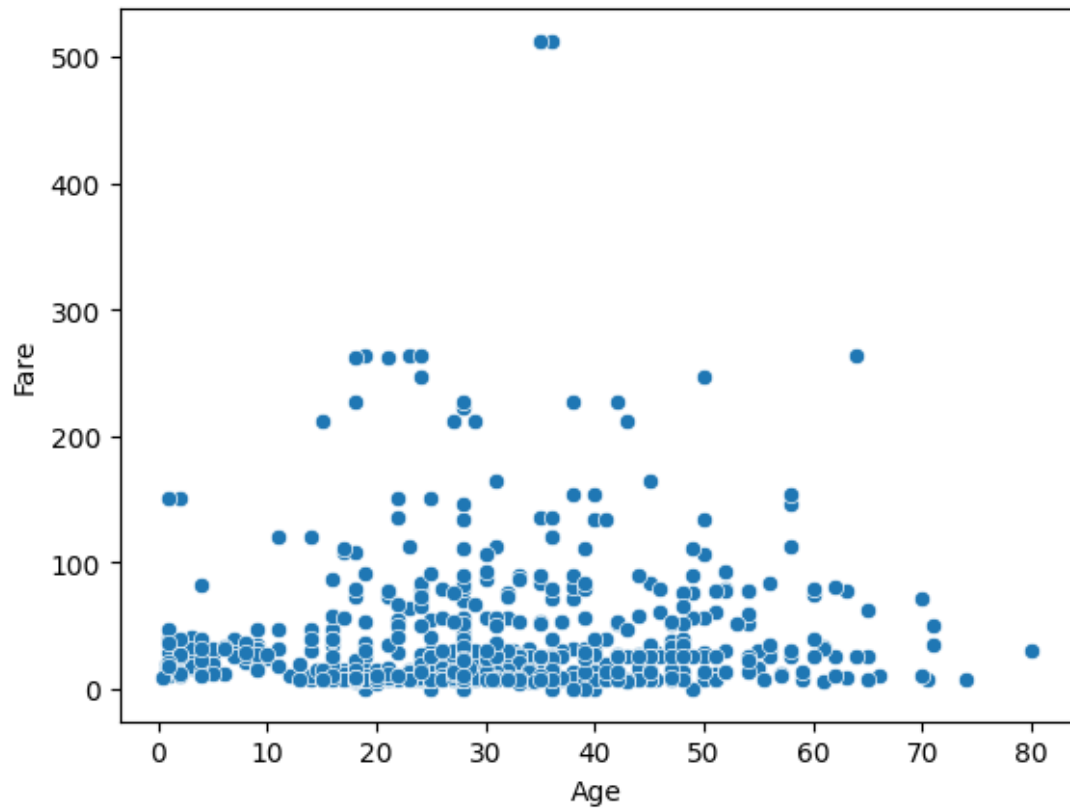
In [30]

```
sns.countplot(x="Sex", hue="Survived", data=df)  
plt.show()
```



In [31]

```
sns.scatterplot(x="Age",y="Fare",data=df)
plt.show()
```



In [32]

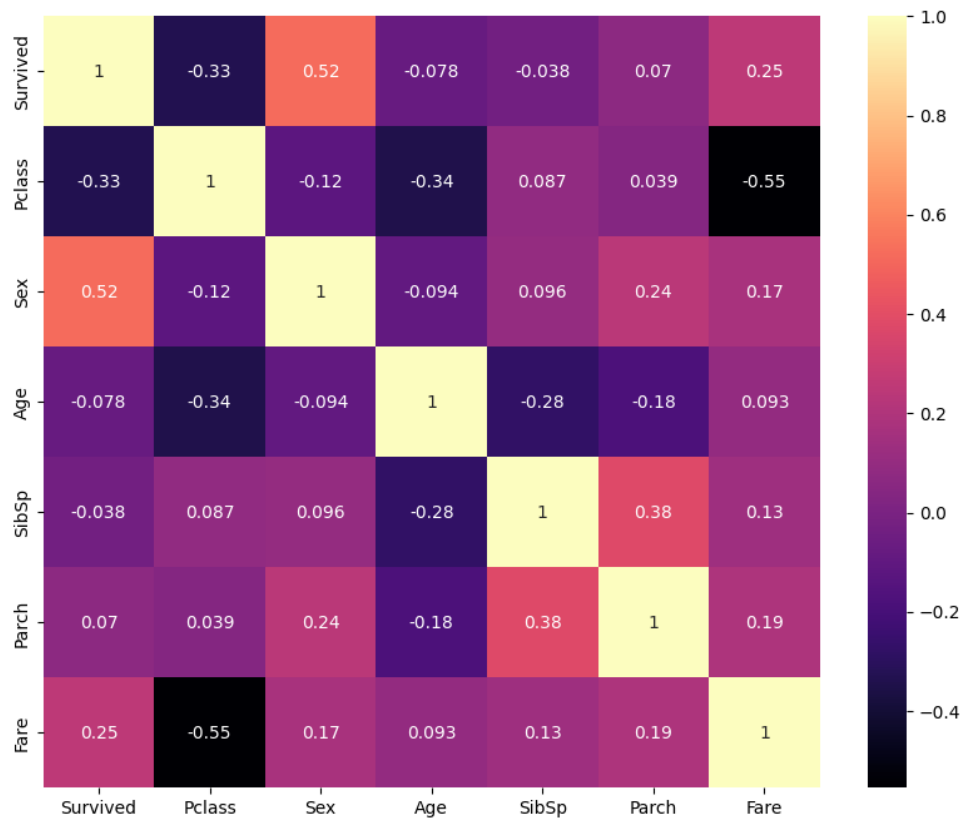
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 8))

numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='magma'
)

plt.show()
```



In [33]

```
x=df.drop("Survived",axis=1)
y=df["Survived"]
```

In [34]

```
x.head(15)
```

Output:

	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked_0.0	Embarked_1.0	\
0	3	0	22.0	1	0	7.2500	True	False	
1	1	1	38.0	1	0	71.2833	False	True	
2	3	1	26.0	0	0	7.9250	True	False	
3	1	1	35.0	1	0	53.1000	True	False	
4	3	0	35.0	0	0	8.0500	True	False	
5	3	0	28.0	0	0	8.4583	False	False	
6	1	0	54.0	0	0	51.8625	True	False	
7	3	0	2.0	3	1	21.0750	True	False	
8	3	1	27.0	0	2	11.1333	True	False	
9	2	1	14.0	1	0	30.0708	False	True	
10	3	1	4.0	1	1	16.7000	True	False	
11	1	1	58.0	0	0	26.5500	True	False	
12	3	0	20.0	0	0	8.0500	True	False	
13	3	0	39.0	1	5	31.2750	True	False	
14	3	1	14.0	0	0	7.8542	True	False	

	Embarked_2.0
0	False
1	False
2	False
3	False
4	False
5	True
6	False
7	False
8	False
9	False
10	False

11	False
12	False
13	False
14	False